

BUSINESS ANALYTICS PROJECT

Business Analytics for Food Delivery Order and Revenue Analysis

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1. Introduction

Business Analytics uses data, statistics, visualization and analytical models to support better business decisions. This project focuses on the food delivery business, where order, restaurant, customer, payment and delivery information can be analyzed to understand revenue and operational performance.

Python was used for data analysis and statistical modelling, while Tableau was used to create the business intelligence dashboard.

2. Problem Statement

To analyze food delivery order and revenue data to identify sales and operational patterns, understand relationships among business variables, evaluate uncertainty, forecast future revenue and recommend strategies for improving business performance.

3. Objectives

32. To collect and analyze food delivery business data.
33. To perform data preprocessing and exploratory data analysis.
34. To identify relationships among order value, delivery time, discount, rating and distance.
35. To analyze revenue and order patterns using Python and Tableau.
36. To apply probability, sampling, confidence intervals and statistical testing.
37. To forecast future revenue using time series techniques.
38. To apply optimization, simulation, ANOVA and experimental design.
39. To derive business insights and recommendations.

4. Business Scenario / Domain Description

The selected domain is online food delivery. The business connects customers with restaurants and manages ordering, payment and delivery operations.

- Services: Online food ordering and delivery.
- Target customers: Digital food-delivery users.
- Operations: Order placement, restaurant processing, payment and delivery.
- Key concerns: Revenue, order volume, delivery time, discounts, ratings and order completion.

5. Dataset Description

Dataset: food_delivery_data.csv. It contains 1,000 records and 12 columns.

Column	Description	Type
Order_ID	Unique order identifier	Categorical
Order_Date	Order date/time	Date/Time
Customer_ID	Customer identifier	Categorical
Restaurant	Restaurant name	Categorical
Food_Category	Food category	Categorical
Order_Value	Order amount	Numerical

Delivery Time	Delivery time in minutes	Numerical
Discount	Discount percentage	Numerical
Rating	Customer rating	Numerical
Payment Method	Payment method	Categorical
Distance	Delivery distance in km	Numerical
Order Status	Completed/Cancelled	Categorical

6. Data Collection

The dataset was prepared as a structured food-delivery business dataset for academic analysis. It contains 1,000 transaction records covering January 2025 to May 2026 and includes sales, restaurant, customer, payment and delivery variables.

7. Data Preprocessing

- CSV loaded using pandas.
- Order_Date converted to datetime64.
- Missing values checked: 0 in every column.
- Duplicate records: 0.
- Data types and numerical ranges checked.
- Final dataset size: 1,000 rows × 12 columns.

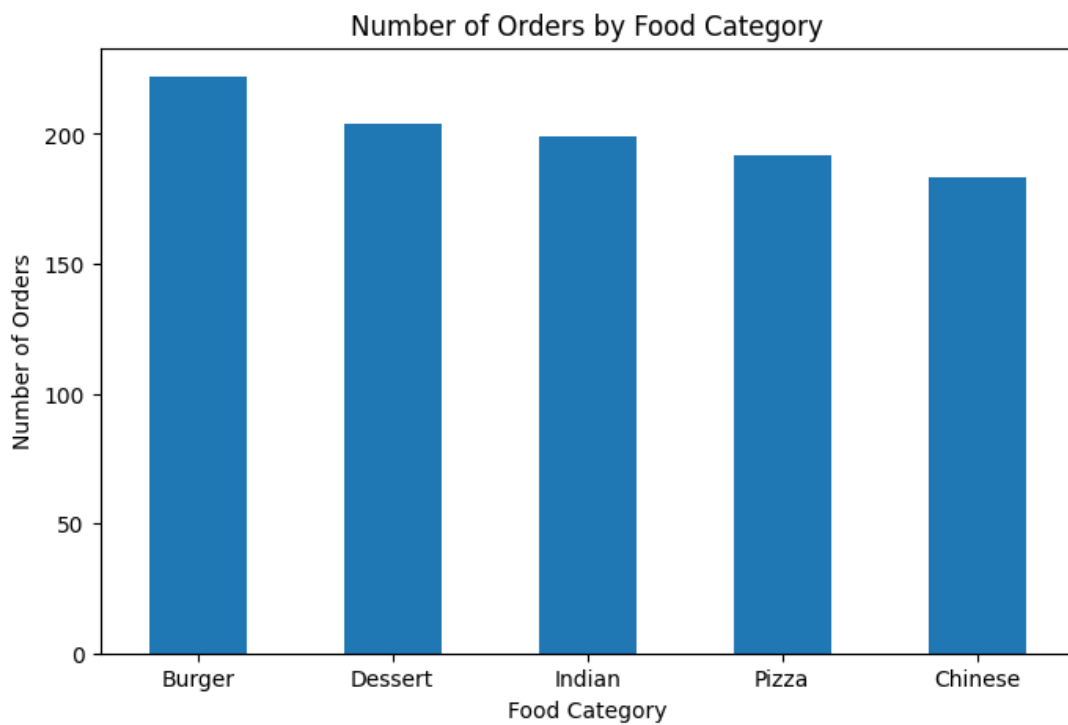
Check	Result
Rows	1000
Columns	12
Missing values	0
Duplicates	0
Order Date	datetime64[ns]

8. Exploratory Data Analysis

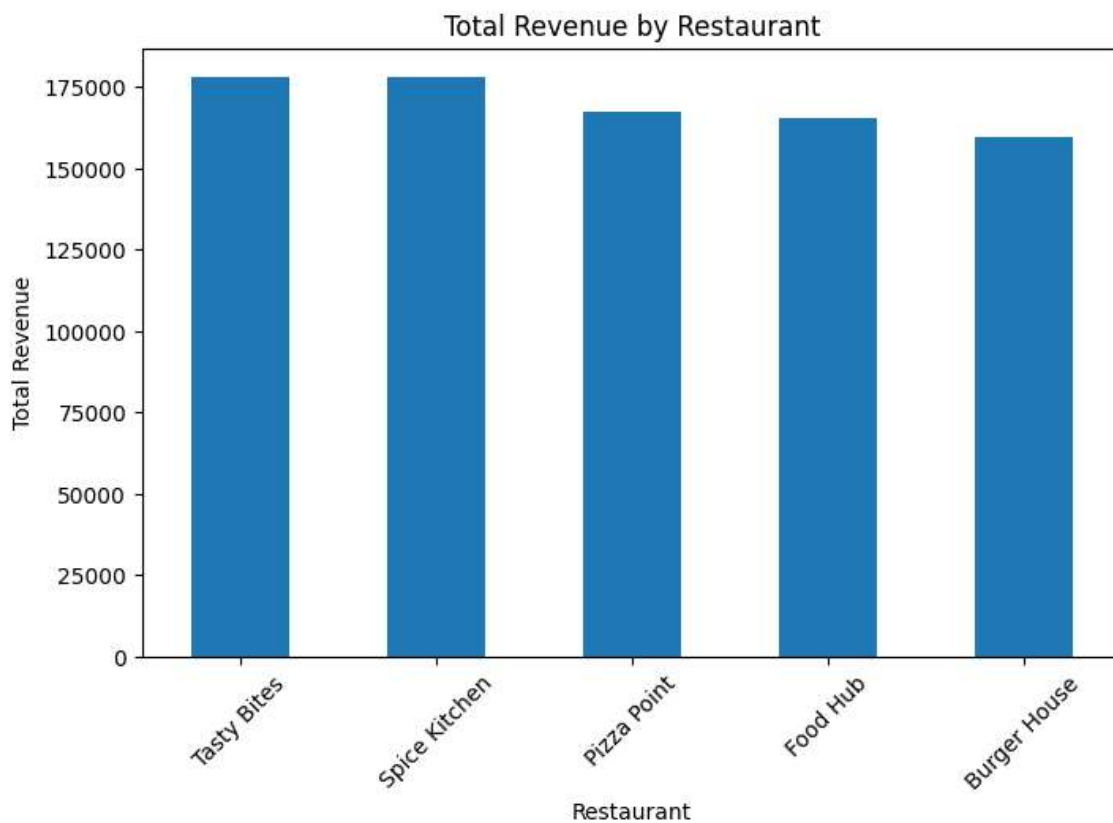
EDA included first/last records, shape, column names, data types, missing values, duplicates, unique values and descriptive statistics.

Measure	Result
Most ordered category	Burger – 222 orders
Highest revenue restaurant	Tasty Bites – ₹177,998.86
Average Order Value	₹848.57
Average Delivery Time	37.58 minutes
Average Rating	3.999

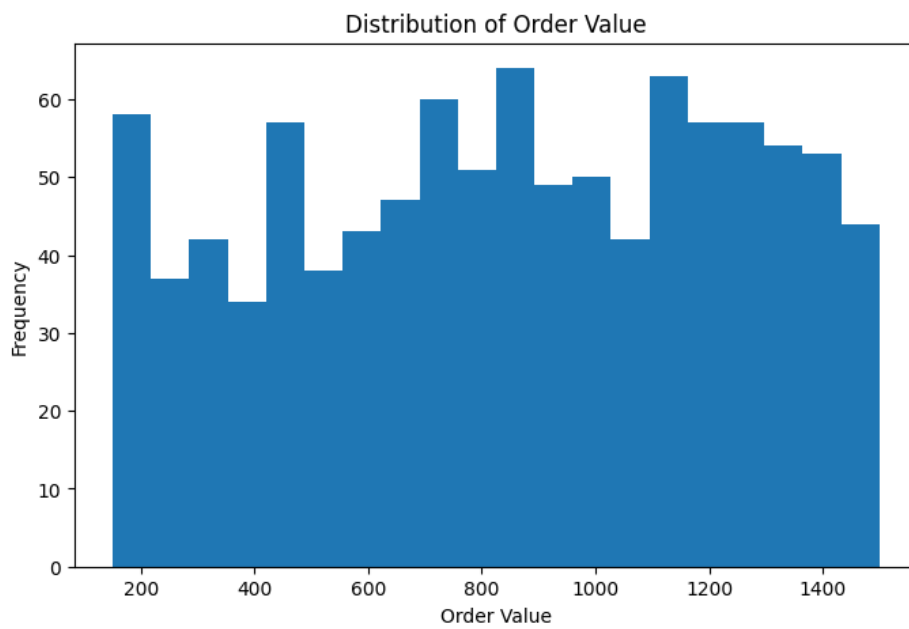
- Orders by Food Category bar chart.



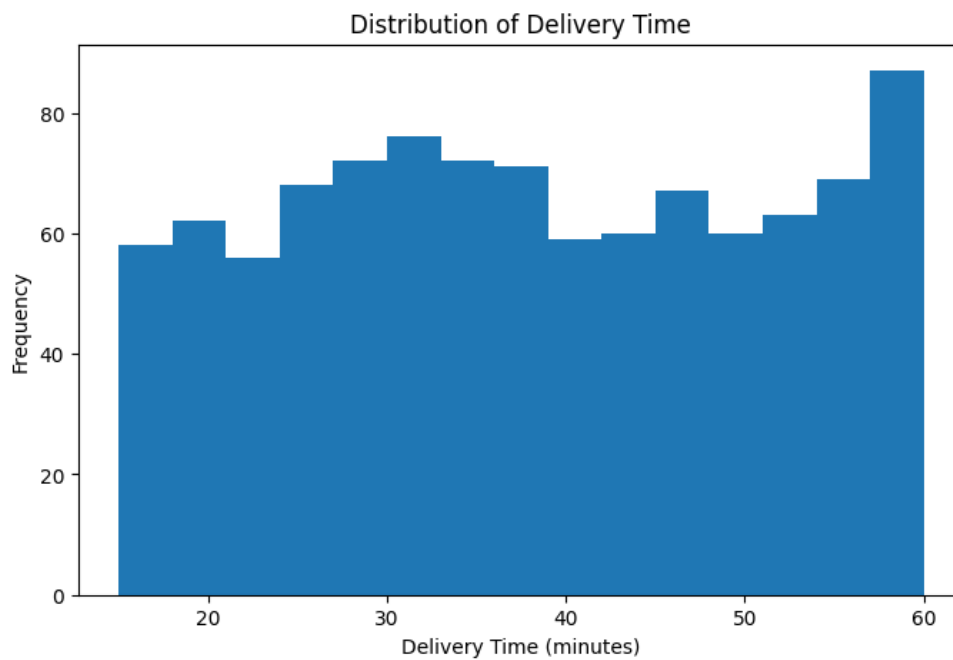
- Revenue by Restaurant bar chart.



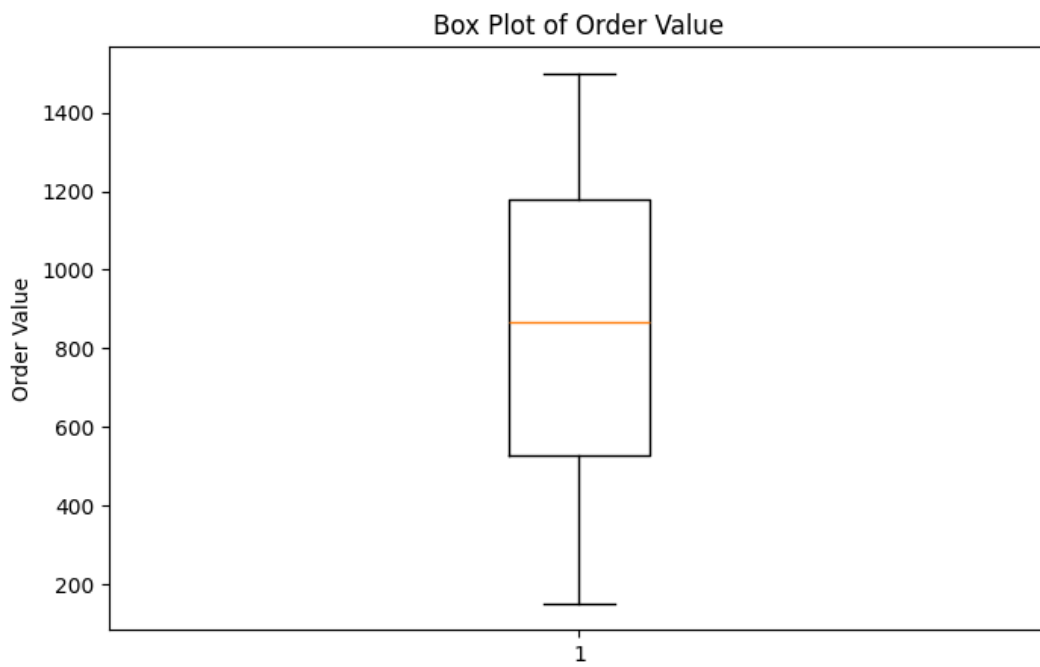
- Order Value histogram.



- Delivery Time histogram.

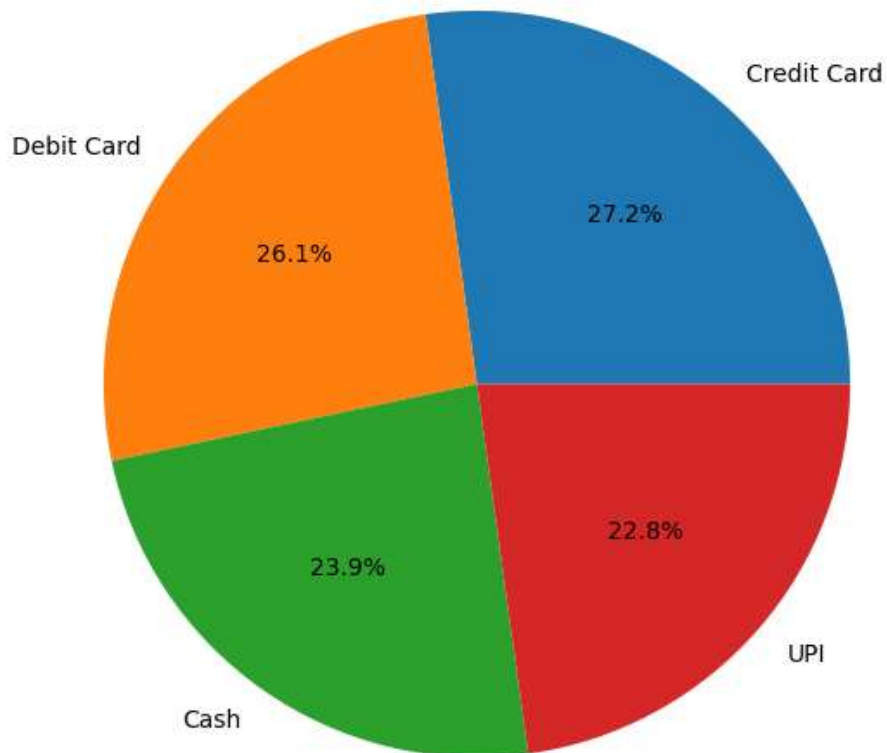


- Order Value box plot.

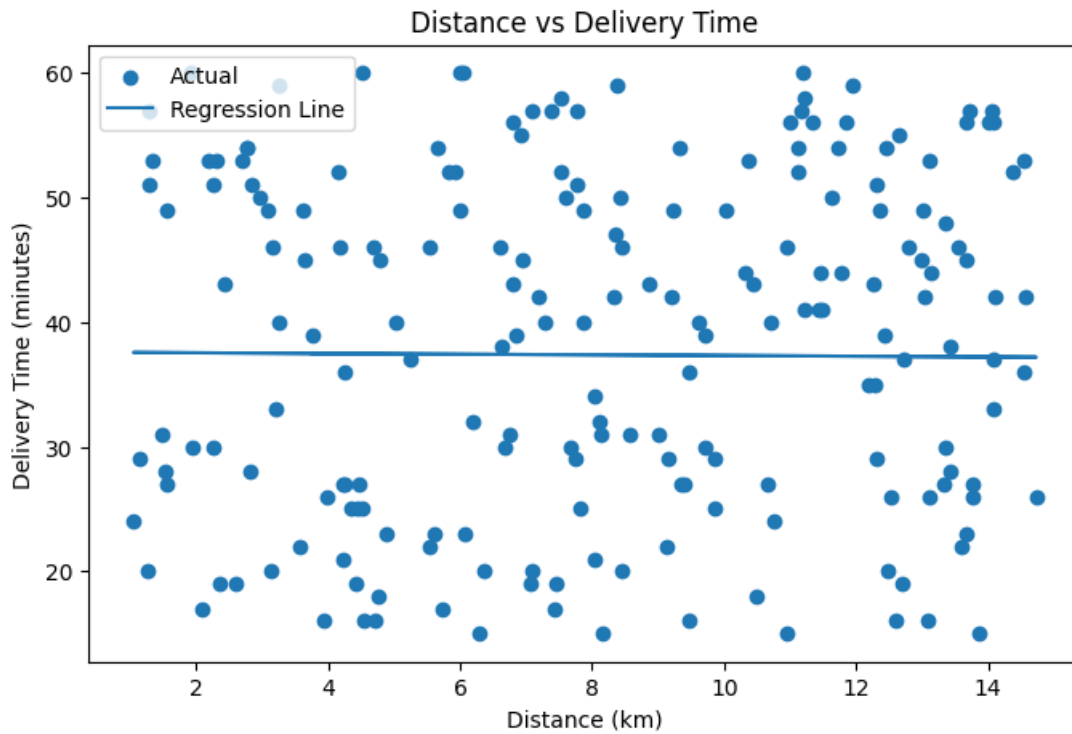


- Payment Method pie chart.

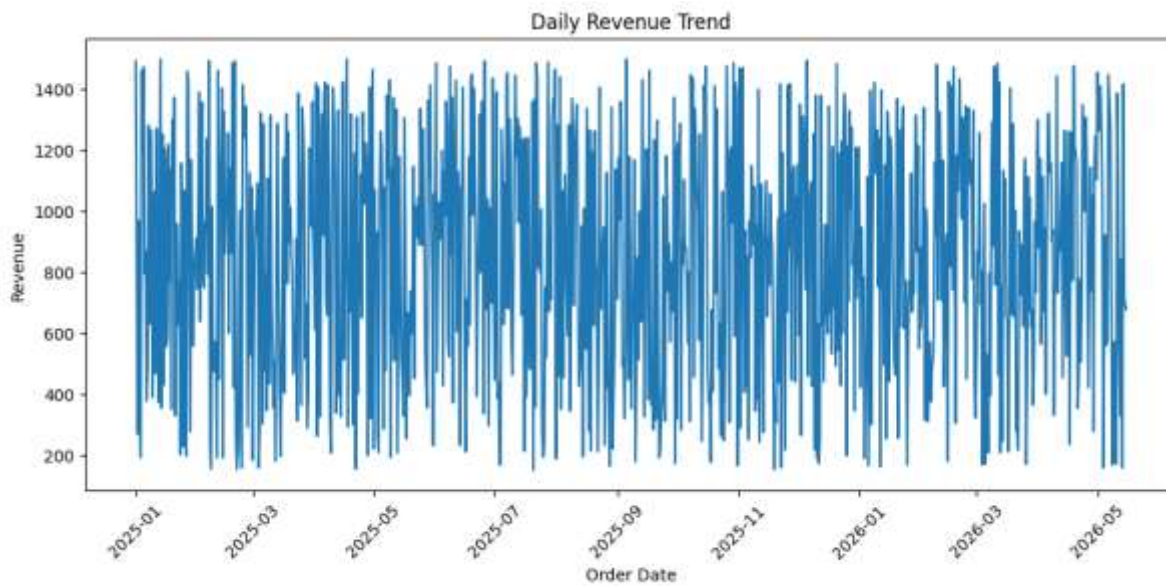
Distribution of Payment Methods



- Distance vs Delivery Time scatter plot.



- Revenue trend line chart.



9. Descriptive Statistical Analysis

Statistic	Order Value
Mean	848.57
Median	867.84
Mode	355.64
Minimum	150.60
Maximum	1499.61
Range	1349.01

Variance	148342.01
Standard Deviation	385.15

The mean order value is ₹848.57 and the median is ₹867.84. The standard deviation of ₹385.15 indicates considerable variation in order values.

10. Distribution Analysis

Order Value was analyzed using a histogram and box plot to study central tendency, spread and possible extreme observations. Normal, uniform and exponential distribution graphs were also generated to illustrate common probability distributions.

11. Correlation Analysis

Variable Pair	Correlation
Order Value – Delivery Time	0.0176
Order Value – Discount	0.0110
Order Value – Rating	0.0422
Order Value – Distance	0.0249
Delivery Time – Distance	0.0113

The relationships are very weak. Delivery Time and Order Value have a correlation of about 0.018, indicating almost no linear relationship.

12. Regression Analysis

Measure	Result
Training records	800
Testing records	200
Intercept	37.59
Coefficient	-0.03
Equation	Delivery Time = 37.59 - 0.03 × Distance
R ²	-0.0076
MSE	184.69
RMSE	13.59
10 km prediction	37.32 minutes

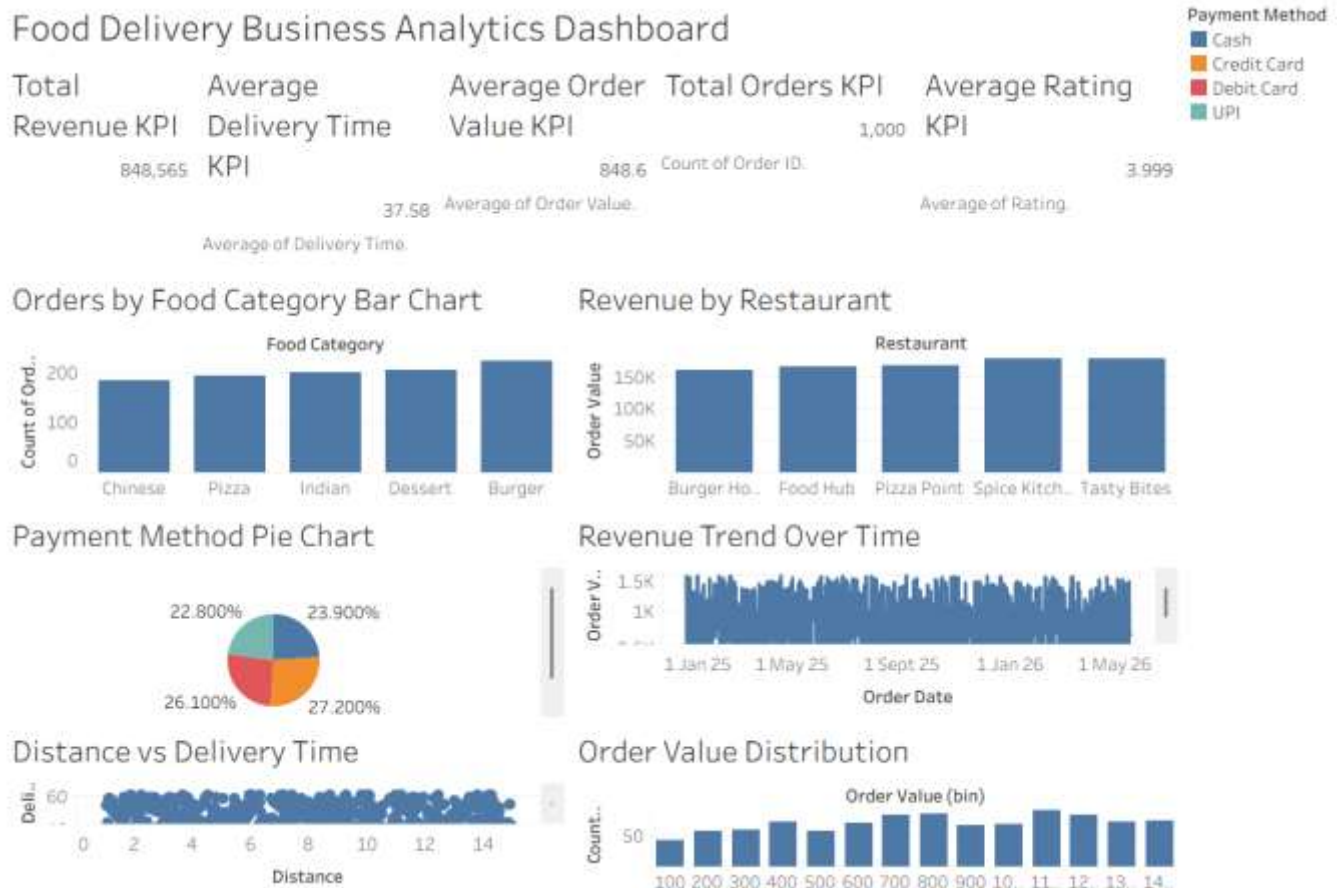
Distance alone has very low predictive power for Delivery Time. Additional operational variables would be required for a stronger model.

13. Business Intelligence Dashboard

A Tableau dashboard was created with KPI cards and visualizations for orders, revenue, order value, delivery time, rating, food categories, restaurants, payment methods, revenue trend, distance and delivery time, and order-value distribution.

- Total Orders
- Total Revenue
- Average Order Value
- Average Delivery Time

- Average Rating
- Orders by Food Category
- Revenue by Restaurant
- Payment Method Distribution
- Revenue Trend Over Time
- Distance vs Delivery Time
- Order Value Distribution



14. Probability and Probability Distributions

Measure	Result
Total Orders	1000
Completed Orders	930
Probability of Completed Order	0.93 (93%)

Normal, uniform and exponential distribution graphs were generated using Python.

15. Decision Making Under Uncertainty

Demand	Probability	Regular Profit	Express Profit
Low	0.30	5000	3000
Medium	0.45	9000	11000
High	0.25	12000	16000
Expected Value		8550	9850

Express Delivery has the higher expected value of ₹9,850 compared with ₹8,550 for Regular Delivery under the stated assumptions.

16. Sampling and Sampling Distribution

Sample Size	Mean of Sample Means	SD	SE
30	864.27	69.65	12.72
50	851.36	52.12	7.37
100	843.85	40.21	4.02

The standard error decreases as sample size increases, demonstrating the sampling distribution and Central Limit Theorem concept.

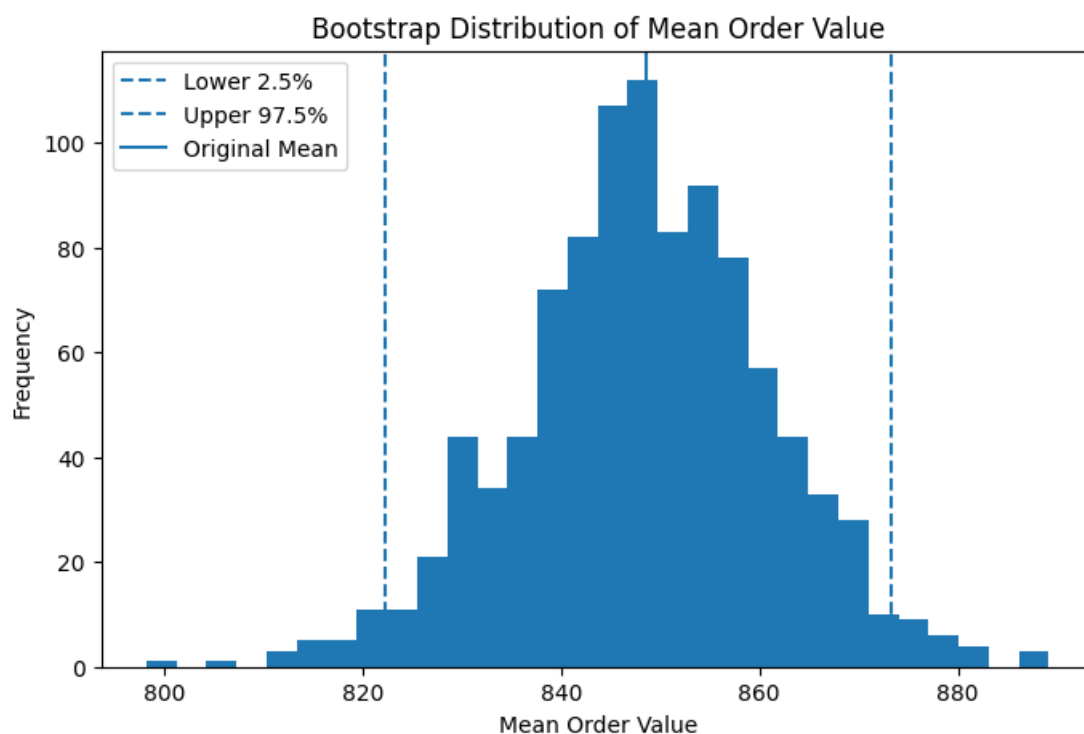
17. Confidence Interval Estimation

Level	Lower	Upper	Margin of Error
90%	828.53	868.60	20.03
95%	824.69	872.44	23.87
99%	817.19	879.94	31.37

The 95% interval estimates the population mean Order Value between ₹824.69 and ₹872.44 under the calculation assumptions.

18. Bootstrapping

A bootstrap analysis used 1,000 samples with replacement from Order Value and calculated the mean for each sample. A bootstrap distribution and 95% bootstrap confidence interval were generated.



19. Hypothesis Testing – Independent Two-Sample t-Test

- **H₀:** The two means are equal.
- **H₁:** The two means are different.
- **Significance level:** $\alpha = 0.05$
- **Test:** Independent two-sample t-test

Results:

- Low Discount mean Order Value = **₹832.27**
- High Discount mean Order Value = **₹856.38**
- **t-statistic = -0.9264**
- **p-value = 0.3545**

Decision:

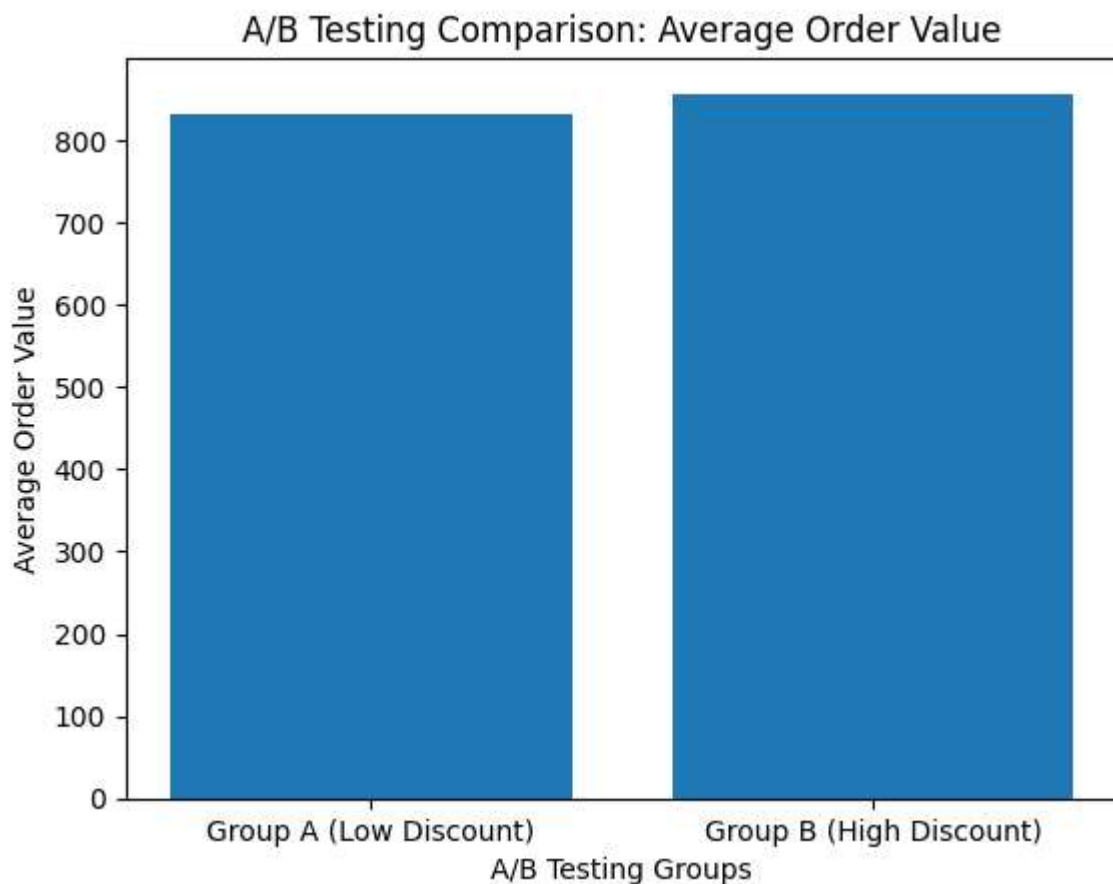
Since **p-value (0.3545) > α (0.05)**, we fail to reject H₀.

Business Interpretation:

There is **no statistically significant difference** in average order value between the Low Discount and High Discount groups. Therefore, based on this test, the discount level does not appear to have a significant effect on average order value.

20. A/B Testing

- Group A: Discount < 10% (Low Discount).
- Group B: Discount $\geq$ 10% (High Discount).
- Response: Order Value.
- H₀: Equal average Order Value.
- H₁: Different average Order Value.
- Significance level: $\alpha = 0.05$.
- Test: Independent two-sample t-test.



21. Time Series Analysis

Order_Date was converted to datetime and daily revenue was calculated. A revenue trend chart and 7-day moving average were created. Exponential Smoothing was then used to forecast the next 30 days.

The project instructions also request ARIMA. The completed analysis used Exponential Smoothing; an ARIMA comparison should be added if required by the faculty.

22. Forecast Accuracy

Measure	Value
MAE	320.07
MSE	140521.09
RMSE	374.86

Lower MAE, MSE and RMSE indicate better forecast accuracy.

23. Optimization Modeling

- Decision variables: x = Regular Deliveries, y = Express Deliveries.
- Objective: Maximize $80x + 120y$.
- Constraint: $2x + 3y \leq 300$.
- Constraint: $x + y \leq 120$.
- Non-negativity: $x \geq 0, y \geq 0$.

- Python method: `scipy.optimize.linprog`.

```
Optimization Result
-----
Optimal Regular Deliveries: 0.0
Optimal Express Deliveries: 100.0
Maximum Profit: 12000.0
```

24. Simulation Modeling

A 30-day demand simulation was created using random demand values between 50 and 150 orders per day. Low, Medium and High Demand scenarios were compared using expected order counts of 80, 100 and 120.

```
30-Day Simulated Demand
-----
Average Simulated Demand: 100.87
Minimum Demand: 51
Maximum Demand: 149

Simulation Results:
```

	Scenario	Expected Orders	Estimated Revenue
0	Low Demand	80	67885.21
1	Medium Demand	100	84856.51
2	High Demand	120	101827.81

25. ANOVA

- Groups: Burger, Chinese, Dessert, Indian and Pizza.
- H0: All group means are equal.
- H1: At least one group mean is different.
- Significance level: $\alpha = 0.05$.
- Test: One-way ANOVA.

```
ANOVA Results
-----
F-statistic: 0.6777
P-value: 0.6075
Decision: Fail to Reject H0
Conclusion: There is no significant difference in average Order Value among Food Categories.
```

26. Experimental Design

Element	Description
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Factor	Discount Level
Levels	Low <10%, High ≥10%
Response	Order Value
Experimental Units	Food delivery orders

A box plot was used to compare Order Value between the two discount groups.

27. Factorial Experiment

- Factor 1: Food Category.
- Factor 2: Discount Level.
- Response: Order Value.
- Main effect of Food Category.
- Main effect of Discount Level.
- Interaction effect.
- Method: Two-way ANOVA with interaction.

Average Order Value by Food Category and Discount Level:

	Food_Category	Discount_Level	Order_Value
0	Burger	High Discount	859.991141
1	Burger	Low Discount	847.167945
2	Chinese	High Discount	851.404776
3	Chinese	Low Discount	834.053469
4	Dessert	High Discount	833.707424
5	Dessert	Low Discount	778.432083
6	Indian	High Discount	886.058071
7	Indian	Low Discount	851.921186
8	Pizza	High Discount	847.820661
9	Pizza	Low Discount	853.972254

28. Business Insights and Findings

40. Burger was the most ordered category with 222 orders.
41. Tasty Bites recorded the highest revenue in the reported restaurant aggregation.
42. Average Order Value was ₹848.57.
43. Average Delivery Time was 37.58 minutes.
44. Average Rating was approximately 4.00.
45. Delivery Time and Order Value had a very weak correlation of 0.018.
46. Distance-only regression had very low predictive performance with $R^2 = -0.0076$.
47. Probability of a completed order was 93% based on 930 completed orders.
48. Express Delivery had higher expected profit: ₹9,850 versus ₹8,550.

49. Sampling standard error decreased from 12.72 at sample size 30 to 4.02 at sample size 100.

29. Business Recommendations

- Promote Burger products due to high order volume.
- Study the practices of the highest-revenue restaurant.
- Use suitable bundles and offers to increase average order value.
- Improve delivery operations to reduce delivery time.
- Maintain service quality and customer ratings.
- Use targeted discounts and evaluate them with A/B testing.
- Use Express Delivery selectively during high-demand situations.
- Use larger samples for important business estimates.

30. Conclusion

This project analyzed food delivery order and revenue data using Python and Tableau. It covered preprocessing, EDA, descriptive statistics, distributions, correlation, regression, dashboarding, probability, decision-making, sampling, confidence intervals, bootstrapping, hypothesis testing, A/B testing, forecasting, forecast accuracy, optimization, simulation, ANOVA and experimental design.

The analysis identified important patterns in orders, revenue, delivery performance and ratings. The Tableau dashboard summarized major KPIs and trends. The results can support decisions related to promotions, delivery operations, customer service, forecasting and resource planning.

31. Appendix

- Complete Python code.
- Dataset sample / CSV.
- EDA charts and outputs.
- Tableau dashboard screenshot.
- Probability distribution graphs.
- Sampling and confidence interval outputs.
- Hypothesis and A/B test outputs.
- Forecast and accuracy graphs.
- Optimization and simulation outputs.
- ANOVA and factorial experiment outputs.